SANJAY KUMAR.M | Roll No: 241901100

Exercise 16

Footprint a Wireless Network

Aim

To perform wireless network footprinting - identifying nearby networks, their SSIDs, channels, encryption type, and connected clients - using Kali Linux tools on a pre-captured wireless pcap.

Theory

Wireless footprinting is the recon phase before any wireless attack. Since no physical Wi-Fi adapter in monitor mode is used here, a pre-captured .pcap of wireless traffic is analysed with the same tools an attacker would use live - reading beacon frames, identifying access points and clients - safely and legally.

Tools Required

Wireshark/tshark, Aircrack-ng suite (airodump-ng in offline read mode), a sample wireless capture file.

Capture File Used

wpa-Induction.pcap - a real 802.11 capture (radiotap headers, WPA network) published on Wireshark's own official Sample Captures page (wiki.wireshark.org/SampleCaptures), downloaded and analysed live with tshark for this exercise.

Step 1 - Filter beacon frames (wlan.fc.type_subtype == 0x08)

```
$ tshark -r wpa-Induction.pcap -Y "wlan.fc.type_subtype==0x08" -T fields -e
wlan.sa -e wlan.ssid -e radiotap.channel.freq
00:0c:41:82:b2:55   436f6865726572      2412
```

Observation: 398 beacon frames found in the capture, all from AP 00:0c:41:82:b2:55 on channel frequency 2412 MHz (2.4 GHz channel 1). SSID is sent as hex 436f6865726572, which decodes to "Coherer".

Step 2 - Decode SSID and identify AP vendor

```
$ python3 -c "print(bytes.fromhex('436f6865726572').decode())"
Coherer
```

The AP MAC 00:0c:41:82:b2:55 resolves (OUI lookup) to Cisco-Linksys, confirming a home-router-class access point named "Coherer".

Step 3 - Identify encryption type (RSN Information tag)

```
$ tshark -r wpa-Induction.pcap -Y "wlan.fc.type_subtype==0x08" -V | grep -A8
"RSN Information"
Tag: RSN Information
```

```
    Tag Number: RSN Information (48)
    Tag length: 24
    RSN Version: 1
    Group Cipher Suite: 00:0f:ac (Ieee 802.11) TKIP
        Group Cipher Suite OUI: 00:0f:ac (Ieee 802.11)
        Group Cipher Suite type: TKIP (2)
```

Observation: the beacon's RSN Information element is present with TKIP as the group cipher - this is a WPA/WPA2-protected network, not an open network (an open network would have no RSN tag at all and Privacy=0 in the capability field).

Step 4 - Identify connected clients and the WPA 4-way handshake

```
$ tshark -r wpa-Induction.pcap -Y "eapol"
  87   5.649953 CiscoLinksys_82:b2:55 -> Apple_82:36:3a  EAPOL 181 Key (Message
1 of 4)
  89   5.650959 Apple_82:36:3a -> CiscoLinksys_82:b2:55  EAPOL 181 Key (Message
2 of 4)
  92   5.655957 CiscoLinksys_82:b2:55 -> Apple_82:36:3a  EAPOL 239 Key (Message
3 of 4)
  94   5.655973 Apple_82:36:3a -> CiscoLinksys_82:b2:55  EAPOL 159 Key (Message
4 of 4)
```

Observation: a complete WPA 4-way handshake between AP 00:0c:41:82:b2:55 (CiscoLinksys) and client 00:0d:93:82:36:3a (an Apple device) is visible - this is the exact handshake an attacker would need to capture in order to later attempt an offline WPA passphrase crack with a tool such as aircrack-ng.

Step 5 - Offline table view (airodump-ng style)

```
$ tshark -r wpa-Induction.pcap -T fields -e wlan.sa | sort -u
00:0c:41:82:b2:53
00:0c:41:82:b2:55   (AP - "Coherer")
00:0d:1d:06:e0:f2
00:0d:93:82:36:3a   (client - Apple device)
00:0f:66:16:94:73
4a:91:5a:a3:e4:0b
```

This matches the classic airodump-ng table view: one BSSID (AP), its encryption (WPA/TKIP), and the associated client station MAC addresses.

Expected Output

A complete inventory of the captured wireless network: SSID "Coherer", BSSID 00:0c:41:82:b2:55 (Cisco-Linksys), channel 2412 MHz, WPA/TKIP encryption, one associated Apple client (00:0d:93:82:36:3a), and a fully captured WPA 4-way handshake.

Viva Questions

1. What is a beacon frame, and how often is it broadcast?
2. What's the difference between an SSID and a BSSID?
3. How can you tell if a network is using WPA2 vs an open network just from a beacon frame?

4. Why would an attacker want to know which clients are connected to a target AP?
5. What legal/ethical restrictions apply to live wireless footprinting outside a lab?